STIVEN BRIAND MASSALA

Email · Google Scholar · GitHub
Generative AI | Machine Learning | Applied Mathematics

EDUCATION

Ph.D. in Deep Learning & Partial Differential Equations *Dec 2022 – June 2026*
Ecole Normale Supérieure & Nanyang Technological University, Singapore
Research: Deep Learning, Self-Supervised Learning, Generative AI, Physics *Magna Cum Laude*

Master of Research in Applied Mathematics *2020 – 2021*
CentraleSupélec, Université Paris-Saclay, France *GPA: 4.0/4.0*
Coursework: Probability & Statistics · PDEs · Linear Algebra · Machine Learning · Inverse Problems

Diplôme d'Ingénieur *2017 – 2021*
Arts & Métiers ParisTech, France
Coursework: Optimization · Control · Fluid Mechanics · Computational Modeling · Scientific Computing

Classes Préparatoires aux Grandes Écoles *2016 – 2017*
Lycée César Baggio, Lille, France *GPA: 4.0/4.0*
Coursework: Maths · Physics · Engineering · Philosophy

PROFESSIONAL EXPERIENCE

Co-founder, Ruptur *June 2026 – Present*
Generative AI & Machine Learning Systems Paris, France

- Lead the development of **Generative AI systems** from research to prototyping.
- Built **RAG pipelines and agentic workflows**, integrating retrieval, tool use.
- Developed domain-specific **LLM post-training pipelines** using Supervised Fine-Tuning (SFT).

Brother Capital *Aug 2026 – Present*
Systematic Trading Paris, France

- Systematic XAU/USD trading strategy.

Machine Learning Researcher, CNRS@CREATE *June 2022 – Apr 2026*
Deep Learning & Physics Singapore

- Conducted applied research in **deep learning and self-supervised learning** for complex, real-world dynamical systems.
- Developed **self-supervised JEPA architectures** for learning representations and predicting time-dependent dynamics in partially known systems.
- Developed **DeepONet and Fourier Neural Operator (FNO)** frameworks for real-time state estimation, reducing simulation latency from hours to seconds.

Research Intern, ANSYS *Apr – Oct 2021*
Computational Physics Lyon, France

- Developed and validated computational models for offshore wind turbine simulations, supporting engineering reliability assessments.

TECHNICAL SKILLS

Generative AI	LLM · RAG · Agentic AI · Tool Use · SFT
Machine Learning	Deep Learning · Self-Supervised Learning · Time Series · Statistical Learning
Frameworks	PyTorch · JAX · TensorFlow · Python
Computing	HPC · Slurm · Linux
Mathematics	Probability · Statistics · PDEs · Numerical Methods
Languages	French (native) · English (fluent)

PUBLICATIONS

Journal Articles

[J2] **Orthogonally Constrained Operator Learning for Bias-Aware Digital Twins: A PBDW–DeepONet Framework**, *Journal of Scientific Computing, Springer Nature*, 2026

[J1] **Joint Embedding Predictive Architectures for Anticipating Model Bias in Real-Time**
Under review, 2026

Conference Papers & Posters

[C8] **From Simulations to Surveys: Domain Adaptation for Galaxy Observations**
NeurIPS — Machine Learning and the Physical Sciences (ML4PS) Workshop, San Diego, USA 2025

[C7] **Real-Time Bias Correction Using Machine Learning and Physics**
MORTech 2025 — VII Int. Workshop on Model Order Techniques, Zaragoza, Spain 2025

[C6] **State, Parameter and Bias Recovery: A Bias-Aware Kalman Filter (BKF)**
ADMOS 2025 — XII Int. Conference on Adaptive Modeling and Simulation, Barcelona, Spain 2025

[C5] **Implementation of Hybrid AI Models for Complex Dynamical Systems of the Smart City**
IACM DTE 2025 & ECCOMAS AICOMAS 2025, Paris, France 2025

[C4] **Hybrid Physics-AI for Efficient Bias-Aware State Estimation**
IAIFI Workshop, MIT, Boston, USA 2024

[C3] **Hybrid Physics-AI for Efficient Bias-Aware State Estimation**
*SINFRA Workshop, NUS-A*STAR*, Singapore 2024

[C2] **Coupling Variational Data Assimilation and Operator Learning for Effective State Estimation on Complex Systems**
16th WCCM & 4th PANACM, Vancouver, Canada 2024

[C1] **Hybrid Twins Coupling PBDW and Machine Learning for Effective State Estimation on Complex Systems**
2nd IACM MMLDE-CSET, El Paso, Texas, USA 2023

WORKSHOPS & SUMMER SCHOOLS

- Mathematical & Statistical Foundations of Data Assimilation — **IMSI**, Chicago 2025
- IAIFI Summer School — **Harvard University**, Boston 2025
- Optimal Control and Decision Making Under Uncertainty — **IMSI**, Chicago 2025
- IAIFI Summer School — **MIT**, Boston 2024
- Symposium on Machine Learning and Dynamical Systems — **Fields Institute** 2024
- Model Reduction & Data Assimilation — **INI, Cambridge University** 2023
- Physics-Informed Neural Networks Summer School — **KTH Royal Institute of Technology** 2023
  Lecturers: Prof. George Karniadakis & Prof. Khemraj Shukla

HONORS & COMPETITIONS

- QRT Stock Prediction Challenge - Rank **64 / 1304 (53% accuracy, predicting the return of a stock in the US market using historical data)**
- International Mathematics Competition 2025 — Rank **31 / 296**
  (*High-Speed Signal Simulation & Equalization*)
- International Mathematics Competition 2023 — Rank **37 / 196**
  Spatial Estimation & Separation